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Five wearable sensors for objective measurement of near work and light exposure

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This study compared five wearable sensors (VEET, Clouclip, Actiwatch, ActLumus, and GENEActiv) for quantification of myopia-related environmental exposures. Thirty participants (27.8 ± 7.0 years) simultaneously wore two spectacle-mounted and three wrist-worn sensors for one day while maintaining a detailed activity log. Data were binned by viewing distance (near: 15– < 60 cm, intermediate: 60– < 100 cm, far: ≥100 cm) and illumination (scotopic: <3 lux, mesopic: 3 to < 30 lux, indoor photopic: 30 to <1000 lux, outdoor photopic: ≥ 1000 lux). The VEET recorded significantly more near and intermediate time than the Clouclip ($p < 0.001$). Activity logs yielded significantly lower near viewing compared to the VEET ($p < 0.001$) and Clouclip ($p = 0.048$) and higher intermediate viewing time ($p < 0.001$). All sensors demonstrated excellent agreement for categorizing outdoor illumination. Activity logs reported significantly greater outdoor durations compared to the VEET, Clouclip, GENEActiv and Actiwatch ($p < 0.05$ for all). The Actiwatch recorded significantly greater scotopic (< 3 lux) durations compared to other sensors ($p < 0.05$ for all). While subjective logs provided useful contextual information, visual activity estimates were significantly different than objective measures. Small but significant discrepancies across sensors emphasize the importance of device selection in myopia research.

Myopia is a growing global public health crisis[1], with projections indicating that nearly half of the world's population will be nearsighted by 2050[2]. The associated rise in high myopia significantly increases the risk of severe, vision-threatening conditions, including myopic macular degeneration, retinal detachment, and glaucoma[3–5]. While genetics play a role, the global surge is believed to be largely driven by environmental and behavioral factors[6]. Extensive evidence points to two primary factors driving myopia, insufficient time outdoors and excessive near work[6–8]. Excessive near work, characterized by prolonged exposure to short viewing distances, is linked to myopia progression through various optical and biomechanical mechanisms[9–11]. Similarly, the protective effects of outdoor time are strongly linked to the intensity and spectral properties of ambient light[12]. High intensity illumination has been shown to stimulate the release of retinal dopamine, a neuromodulator known to protect against experimental myopia[13]. Crucially, retinal dopamine exhibits a strong diurnal rhythm intrinsically linked to the circadian system[14], highlighting a shared mechanistic dependence on daily environmental light exposure. Consequently, the intersection of myopia and circadian biology is an area of growing interest[15,16].

Historically, studies investigating environmental and behavioral risk factors have relied on subjective methods, such as questionnaires and diaries[17–19]. These approaches suffer from recall bias[18] and cannot continuously track dynamic changes in light exposure or precise viewing distances. More recently, researchers have increasingly adopted wearable technology to capture objective, high-resolution data in real-world settings[20,21]. The objective measurement of individual light exposure is not only foundational to understanding myopiogenesis but is becoming highly relevant to broader circadian and sleep research[22]. Because light is the primary connecting factor, accurately quantifying individual light exposure profiles is essential across these interconnected disciplines. However, a major challenge is device availability and standardization. Many sensors utilized in earlier studies were custom-built, restricted to small groups of researchers, or not commercially available, thus limiting their widespread adoption[23–25]. Compounding this issue, some more commonly used wearable sensors, such as the Actiwatch, are no longer manufactured. As researchers must transition to newer technologies, there is a need for a comprehensive evaluation. Although modern devices may have been validated against laboratory standards individually in separate studies[21,26,27],

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simultaneous continuous real-world comparisons are essential to ensure data consistency, mutual agreement, and comparability.

Significant differences exist among currently available wearable devices, particularly regarding their anatomical placement and underlying technology[20]. Wrist-worn devices, such as the Actiwatch, ActLumus, and GENEActiv, are commonly used for continuous monitoring of ambient light[28,29]; however, their position on the wrist makes them susceptible to obstruction by clothing. In addition, because these devices are not positioned at eye level and are subject to rotational movement, they may provide inaccurate estimates of the illuminance reaching the eye. Spectacle-mounted light sensors, such as the Visual Environment Evaluation Tool (VEET) and Clouclip, are positioned at eye level and are more closely aligned with the line of sight. Both the VEET and Clouclip also incorporate time-of-flight (ToF) technology to estimate viewing distance. To account for typical eye position during near work, these sensors feature a fixed tilt (10° downward for the Clouclip and 20° downward and 4° inward for the VEET). This dual capability enables simultaneous assessment of ambient light exposure and near-work behavior, providing a more comprehensive characterization of visual experience.

The primary objective of this study was to systematically evaluate five wearable sensors to support objective quantification of myopia-related environmental factors. We assess the mutual agreement of these sensors, rather than their absolute validation or reliability, as no true gold standard currently exists for continuous, real-world monitoring of viewing behavior. Viewing distance was compared between two spectacle-mounted devices equipped with ToF technology, the VEET and the Clouclip, and illumination was compared across five devices, the VEET and Clouclip, as well as three wrist-worn devices, the Actiwatch, ActLumus, and GENEActiv. Additionally, self-reported daily activity logs facilitated direct comparisons of categorized task-based viewing and illumination against objective data. By comparing output of these devices, as well as their alignment with subjective logs, this research aims to guide the selection of appropriate instrumentation for future myopia research and intervention trials.

Results

Participant characteristics

Thirty participants (12 males, 18 females) with a mean ± standard deviation (SD) age of 27.8 ± 7.0 years (range: 19–49 years) completed the study. Baseline SER did not differ significantly between the right and left eyes ($p > 0.05$). Based on right eye SER, 12 participants were classified as non-myopes (mean ± SD: +0.10 ± 0.28 D) and 18 participants were classified as myopes (mean ± SD: −2.56 ± 1.90 D).

Wear time

According to participant logs, the median (Q1, Q3) total wear time of the five devices was 14.0 h (13.63, 14.35). The objectively recorded median (Q1, Q3) total wear time, calculated from the VEET was 14.03 h (13.62, 14.18).

Table 1 | Valid data durations and exclusions for viewing distance.

Metric	Clouclip	VEET
Participants with valid data (n)	28	30
Valid hours before exclusion, median (Q1, Q3)	13.72 (12.47, 14.07)	13.12 (11.21, 14.07)
Reason for exclusion	Missing device data ($n = 2$)	Excluded to maintain paired dataset ($n = 2$)
Participants included in paired analysis (n)	28	28
Valid hours after alignment, Median (Q1, Q3)*	13.12 (11.12, 14.07)	13.12 (11.12, 14.07)

*Valid hours after alignment represents the finalized, synchronized duration used for direct point-to-point comparison. To maintain perfect temporal alignment, any transient periods where one device failed to record were subsequently excluded from all paired datasets.

The difference between the two was statistically and clinically non-significant (Wilcoxon signed-rank test, $p = 0.67$). After accounting for intermittent non-wear periods and technical data gaps, the median (Q1, Q3) total wear time for the VEET across 30 participants was 13.12 h (11.21, 14.07). The Clouclip failed to collect data for two participants; thus, the comparative analysis for viewing distance was conducted among 28 participants, for which the median (Q1, Q3) duration of valid data was 13.72 h (12.47, 14.07). The Actiwatch, ActLumus, and GENEActiv each had data for 29 participants, with median (Q1, Q3) valid durations of 13.20 h (12.02,14.07), 13.65 h (12.84, 14.07), and 13.78 h (12.92, 14.07), respectively. For direct comparisons across sensors, only time periods when all sensors had valid data were included. To maintain perfect temporal alignment, any transient periods where a device failed to record were excluded from paired datasets. Consequently, the final comparative analysis included 28 participants for distance with median (Q1, Q3) duration of 13.12 hours (11.12, 14.07) and 25 participants for ambient illumination with median (Q1, Q3) duration of 13.09 h, (10.47, 14.06). Detailed information regarding the number of participants with valid data, median (Q1, Q3) valid wear times before and after exclusion, and the primary reasons for data exclusion are provided in Tables 1, 2 for viewing distance and for ambient illumination, respectively.

Activity logs

Table 3 shows a representative daily activity log from one participant, along with the categorization of these activities into distance and light classifications, which were utilized for subsequent comparisons with objective data. Following the categorization, the total duration for each category was computed. According to the activity log, this participant spent 11.5 h indoors and 2.5 h outdoors. For distance categories, time spent in near work was 4.33 h, in intermediate work was 7.17 h, and in far activities was 2.5 h.

Activity log data across the entire cohort, as well as sensor data, are shown in Table 4. The median (Q1, Q3) self-reported time was 11.32 h (9.88, 12.33) indoors and 1.83 h (1.13, 2.52) outdoors. Similarly, for viewing distance, the self-reported time spent in near viewing was 1.91 h (0.85, 3.78), intermediate viewing was 7.85 h (4.61, 9.92), and in far viewing was 1.96 h (1.06, 2.78).

Viewing distance

Figure 1 shows a representative viewing distance profile from a single participant over the course of the observation period for the VEET and Clouclip. To avoid selection bias, this participant was selected randomly from the subset of participants who maintained 100% complete data capture and optimal signal quality across all devices simultaneously. This participant had 11.7 h of valid wear time, with no non-wear periods or technical data gaps. Although both datasets were temporally aligned and resampled to a uniform 1-second interval, the VEET dataset visually demonstrates greater signal continuity and higher moment-to-moment variance. Furthermore, the VEET captured a wider dynamic range of viewing distances, also recording targets positioned well beyond 100 cm. In contrast, the Clouclip data exhibited a distinct ceiling effect, with maximum distance measurements capped near 100 cm and a sparser overall signal. The gaps observed in the Clouclip data correspond to distances outside its detection range and interpreted as far viewing. Consequently, while both devices effectively detected near and intermediate viewing distances, their absolute tracking capabilities differed notably across the dynamic distance spectrum.

The cumulative viewing durations across the three distance categories, near, intermediate, and far, were compared between the VEET, the Clouclip, and the subjective activity logs (Fig. 2). A Friedman test revealed statistically significant differences among the three measurement methods with large effect sizes for all distance categories: near viewing ($\chi^2(2) = 36.64$, $n = 28$, $p < 0.001$, $W = 0.65$), intermediate viewing ($\chi^2(2) = 52.29$, $n = 28$, $p < 0.001$, $W = 0.93$), and far viewing ($\chi^2(2) = 40.79$, $n = 28$, $p < 0.001$, $W = 0.73$). Post-hoc pairwise comparisons with Dunn–Bonferroni correction (reporting adjusted p-value and effect size r) demonstrated that the VEET captured significantly more time spent in both near (10 to < 60 cm)

Table 2 | Valid data durations and exclusions for illumination.

Metric	Actiwatch	ActLumus	GENEActiv	Clouclip	VEET
Participants with valid data (n)	29	29	29	28	30
Valid hours before exclusion, median (Q1, Q3)	13.20 (12.02, 14.07)	13.65 (12.84, 14.07)	13.78 (12.92, 14.07)	13.72 (12.47, 14.07)	13.12 (11.21, 14.07)
Reason for exclusion	Missing device data ($n = 1$)	Missing device data ($n = 1$)	Missing device data ($n = 1$)	Missing device data ($n = 2$)	Excluded to maintain paired dataset ($n = 5$)
Participants included in paired analysis (n)	25	25	25	25	25
Valid hours after alignment, median (Q1, Q3)*	13.09 (10.47, 14.06)	13.09 (10.47, 14.06)	13.09 (10.47, 14.06)	13.09 (10.47, 14.06)	13.09 (10.47, 14.06)

*Valid hours after alignment represents the finalized, synchronized duration used for direct point-to-point comparison. To maintain perfect temporal alignment, any transient periods where one device failed to record were subsequently excluded from all paired datasets.

Table 3 | Representative daily activity log from one participant.

Time	Activity	Distance classification	Illumination classification
9:00 am–9:45 am	Prepared and ate breakfast	Intermediate	Indoors
9:45 am–10:00 am	Cleaned kitchen	Intermediate	Indoors
10:00 am–12:30 pm	Did laundry and cleaned apartment	Intermediate	Indoors
12:30 pm–1:00 pm	Read and organized papers	Near	Indoors
1:00 pm–1:20 pm	Folded clothes	Intermediate	Indoors
1:20 pm–1:30 pm	Ate snack and used phone	Near + Intermediate	Indoors
1:30 pm–2:30 pm	Used phone	Near	Indoors
2:30 pm–3:00 pm	Cleaned closet and laundry room	Intermediate	Indoors
3:00 pm–3:30 pm	Styled hair	Intermediate	Indoors
3:30 pm–4:00 pm	Wrote letter and packed a bag	Near + Intermediate	Indoors
4:00 pm–5:45 pm	Driving the car	Far	Outdoors
5:45 pm–7:15 pm	Ate dinner and watched movie	Intermediate	Indoors
7:15 pm–8:00 pm	Used phone	Near	Indoors
8:00 pm–8:30 pm	Cleaned kitchen	Intermediate	Indoors
8:30 pm–9:15 pm	Driving the car	Far	Outdoors
9:15 pm–10:15 pm	Used phone	Near	Indoors
10:15 pm–11:00 pm	Read book	Near	Indoors

Self-reported activities were assigned to spatial category (near, intermediate, or far) and ambient illumination was classified as indoors or outdoors.

Table 4 | Median (Q1, Q3) duration (hours) of viewing distance categories and ambient illumination exposure across activity logs and the five sensors utilized in the current study ($N = 28$ for distance; $N = 25$ for illumination).

	Activity Log	VEET	Clouclip	Actiwatch	ActLumus	GENEActiv
Distance						
Near (10–< 60 cm)	1.91 (0.85, 3.78)	6.88 (6.07,7.97)	4.42 (3.07, 6.50)	n/a	n/a	n/a
Intermediate (60–< 100 cm)	7.85 (4.61, 9.92)	2.65 (1.40, 3.51)	1.32 (0.77, 2.20)	n/a	n/a	n/a
Far (≥ 100 cm)	1.96 (1.06, 2.78)	3.03 (1.81, 3.54)	6.10 (4.22, 7.66)	n/a	n/a	n/a
Location						
Indoors	11.32 (9.88, 12.33)	12.57 (10.26, 13.39)	11.57 (9.45, 12.91)	12.23 (10.02, 13.30)	12.45 (10.19, 13.25)	12.60 (10.28, 13.46)
Outdoors	1.83 (1.13, 2.52)	0.46 (0.18, 0.77)	0.50 (0.22, 77)	0.44 (0.18, 0.91)	0.55 (0.23, 0.89)	0.42 (0.16, 0.73)
Illumination						
Scotopic	n/a	0.47 (0.15, 1.13)	0.24 (0.08, 0.65)	3.85 (2.18, 6.22)	0.50 (0.13, 0.76)	0.80 (0.33, 1.84)
Mesopic	n/a	3.71 (2.14, 5.63)	2.91 (1.39, 5.40)	2.46 (0.92, 4.05)	2.17 (0.89, 3.88)	3.01 (1.76, 4.84)
Indoor Photopic	n/a	6.92 (4.18, 9.17)	6.51 (3.67, 9.56)	3.72 (2.36, 5.11)	7.53 (4.83, 10.14)	6.10 (3.97, 9.32)
Outdoor Photopic	n/a	0.46 (0.18, 0.77)	0.50 (0.22, 77)	0.44 (0.18, 0.91)	0.55 (0.23, 0.89)	0.42 (0.16, 0.73)

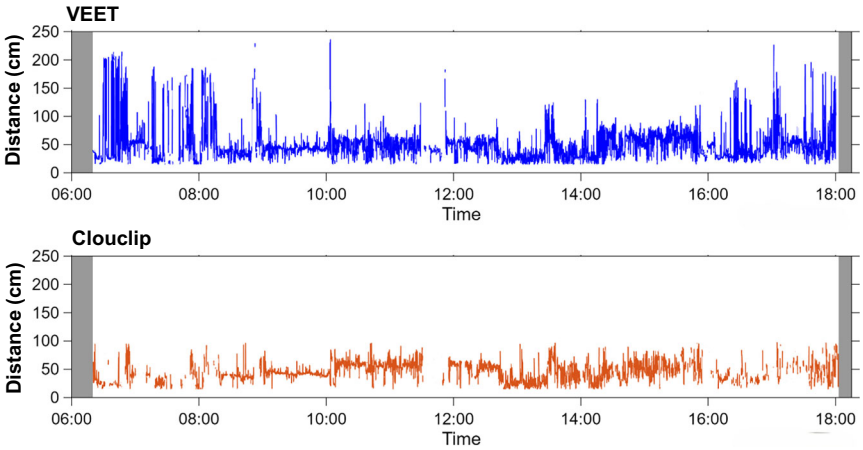


Fig. 1 | Representative viewing distance (cm) from one participant over the observation period for the VEET (top trace, blue) and the Clouclip (bottom trace, orange). The grey shaded regions indicate periods before and after the observation period. The representative participant was selected randomly from the subset of participants who maintained 100% complete data capture and optimal signal quality (zero missing data, wear-time non-compliance, or sensor artifacts) across all wearable devices simultaneously.

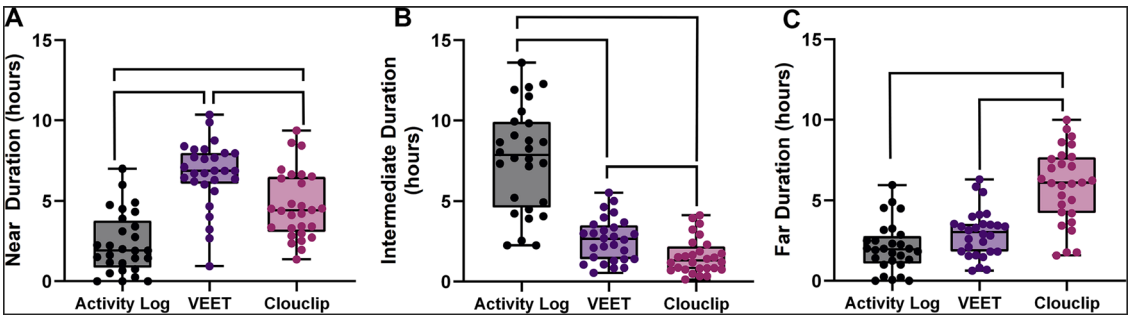


Fig. 2 | Median viewing durations for the activity log, VEET, and Clouclip ($N = 28$). Data are shown for (**A**) near (10 to < 60 cm), (**B**) intermediate (60 to < 100 cm), and (**C**) far (≥ 100 cm). Middle horizontal lines inside the boxes represent the median, and the upper and lower boundaries of the boxes represent the interquartile range. Whiskers represent the minimum and maximum values (range), with all individual data points superimposed as filled circles. Brackets indicate a statistically significant difference between pairs ($p < 0.05$).

($p < 0.001$, $r = 0.68$) and intermediate (60 to < 100 cm) ($p = 0.004$, $r = 0.61$) viewing tasks compared to the Clouclip. Conversely, the Clouclip recorded significantly higher duration than the VEET for far viewing (≥ 100 cm) ($p < 0.001$, $r = 0.78$).

The durations derived from activity logs exhibited significant differences compared to the objective data. The activity logs yielded significantly lower near viewing duration compared to both the VEET ($p < 0.001$, $r > 0.99$) and the Clouclip ($p = 0.048$, $r = 0.45$). In contrast, durations categorized as intermediate viewing from the logs were significantly higher than those recorded by both devices ($p < 0.001$, $r = 0.76$ for VEET and $p < 0.001$, $r > 0.99$ for Clouclip). For far viewing, the log-derived durations were significantly lower than those recorded by the Clouclip ($p < 0.001$, $r > 0.99$), but not different than the VEET ($p = 0.098$, $r = 0.40$). Detailed pairwise test statistics, adjusted p-values, and effect sizes (r) for analysis of each viewing distance category are provided in Supplementary Table S1.

Evaluation of viewing durations between the VEET and Clouclip revealed varying degrees of correlation and notable systematic biases across distance categories (Fig. 3). For near viewing, the devices demonstrated a moderate linear correlation $R^2 = 0.56$, ($p < 0.0001$); however, Bland–Altman analysis confirmed that Clouclip recorded systematically lower near duration relative to VEET, with a mean bias of +2.0 h (95% LoA, −1.0 to +4.9 h). Similarly, intermediate viewing showed a weaker correlation, $R^2 = 0.35$ ($p = 0.001$), with a positive mean bias of +1.0 h; 95% LoA, −1.2 to +3.2 h. Conversely, far viewing exhibited a strong correlation, $R^2 = 0.78$ ($p < 0.0001$), but with substantially higher viewing duration by the Clouclip relative to the VEET, resulting in a negative mean bias of −3.0 h (95% LoA, −5.4 to −0.6 h).

Ambient illumination

Figure 4 shows representative ambient illumination data from one participant across the observation period. To avoid selection bias, this participant was selected randomly from the subset of participants who maintained 100% complete data capture and optimal signal quality across all devices simultaneously. Qualitatively, all five sensors captured major environmental transitions, demonstrating similar timing of peaks and troughs. Short-term fluctuations and intermittent variability were observed across devices. Despite differences in position, the overall dynamic range and temporal patterns were generally consistent between spectacle-mounted and wrist-worn devices, indicating that location did not substantially limit the detection of broad ambient light trends.

Illuminance was categorized into two broad categories, indoors (< 1000 lux) and outdoors (≥ 1000 lux), and compared across the activity log and sensors using Friedman tests (Fig. 5). Both categories revealed statistically significant differences across the measurement methods: indoors ($\chi^2(5) = 51.05, n = 25, p < 0.001, W = 0.41$) and outdoors ($\chi^2(5) = 27.18, n = 25, p < 0.001, W = 0.22$).

Post-hoc Dunn–Bonferroni comparisons (reporting adjusted p-value and effect size r) indicated that for indoors, the activity log duration was significantly less than all durations derived from the sensors (Actiwatch: $p = 0.006, r = 0.71$; ActLumus: $p = 0.017, r = 0.65$; GENEActiv: $p < 0.001, r > 0.99$; VEET: $p < 0.001, r > 0.99$), with the exception of the Clouclip ($p > 0.99, r = 0.26$). Significant differences also existed between some sensors; notably, the Clouclip recorded significantly lower indoor durations compared to both the VEET ($p < 0.001, r = 0.87$) and the GENEActiv ($p < 0.001, r = 0.85$). On the other hand, the activity log duration for outdoors was significantly greater than for all sensors, except for the ActLumus. Detailed pairwise test statistics adjusted p-values, and

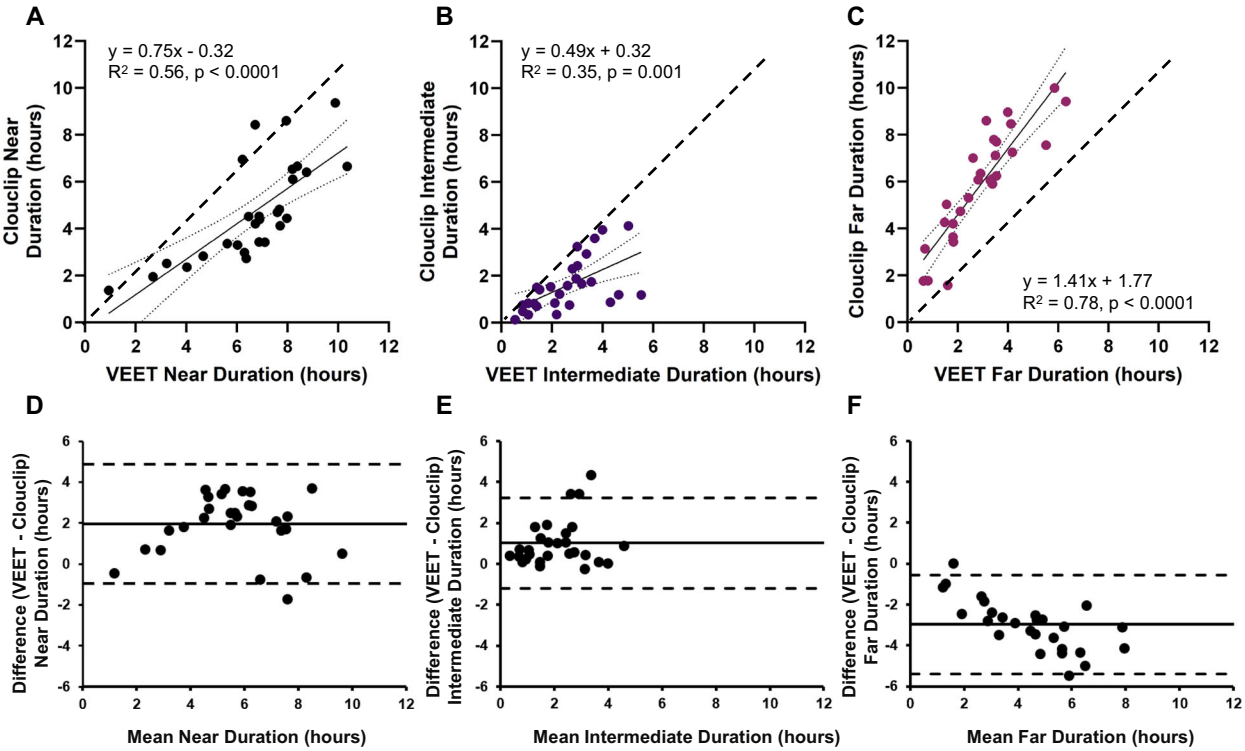


Fig. 3 | Comparison of viewing durations between the VEET and Clouclip across viewing distance categories ($N = 28$). Linear regression (top row) and corresponding Bland–Altman plots (bottom row) are shown for (**A**, **D**) near (10 to < 60 cm), (**B**, **E**) intermediate (60 to < 100 cm), and (**C**, **F**) far (≥ 100 cm) viewing distances. In the regression plots (**A**–**C**), solid lines indicate the linear fit with shaded 95% confidence intervals, and dashed lines represent the identity line ($y = x$). In the Bland–Altman plots (**D**–**F**), solid lines represent the mean bias, and dashed lines represent the 95% limits of agreement (mean bias ± 1.96 × SD).

effect sizes (r) for analysis of each illumination category are provided in Suppl. Table S2.

To compare exposure duration across a range of illuminations, light level was binned into four categories: scotopic (< 3 lux), mesopic (3 to < 30 lux), indoor photopic (30 to < 1000 lux), and outdoor photopic (≥ 1000 lux, Fig. 6). Friedman tests [scotopic: $\chi^2(4) = 75.81, n = 25, p < 0.001, W = 0.76$; mesopic: $\chi^2(4) = 33.63, n = 25, p < 0.001, W = 0.34$; indoor photopic: $\chi^2(4) = 48.35, n = 25, p < 0.001, W = 0.48$; outdoor photopic: $\chi^2(4) = 6.02, n = 25, p = 0.198, W = 0.06$] and post-hoc Dunn–Bonferroni pairwise comparisons revealed that all five sensors demonstrated the strongest agreement in duration of outdoor photopic light exposure (≥ 1000 lux), with no significant differences across devices ($p > 0.05$). Furthermore, the ActLumus, GENEActiv, Clouclip, and VEET showed broad agreement across both the mesopic and indoor photopic illumination ranges. In contrast, the Actiwatch recorded significantly greater durations in the lowest illuminance bin (scotopic) and significantly less time in the typical indoor photopic bin compared to the other four devices. Detailed pairwise test statistics, adjusted pairwise p-values, and effect sizes (r) for each illumination category are provided in Suppl. Table S2.

Discussion

The purpose of this study was to evaluate agreement among five different wearable sensors, the VEET, Clouclip, Actiwatch, ActLumus, and GENEActiv, and subjective activity logs for quantifying viewing distance and ambient light exposure, two environmental factors strongly associated with myopia development and progression[6–8]. Objective monitoring of visual behavior is increasingly utilized in myopia research[20,21,28,29], yet the rapid evolution of wearable technology has created challenges in standardization and cross-study comparability. The results demonstrate that differences exist across currently available devices, particularly for viewing distance. Furthermore, subjective activity estimates were significantly different than objective data, as previously demonstrated[18,30]. These findings highlight the importance of understanding method and device-specific characteristics when interpreting environmental exposure data.

For viewing distance, significant differences were observed between the VEET and Clouclip across all distance categories. The VEET recorded greater durations of near and intermediate viewing, whereas the Clouclip recorded greater durations of far viewing. These discrepancies are likely attributable to differences in sensor parameters and positioning. Although both devices utilize time-of-flight (ToF) technology and have previously demonstrated acceptable laboratory accuracy[21,27], their optical geometry, beam width, sampling strategy, and detection range differ substantially. The Clouclip projects a narrower infrared beam and returns a single distance estimate, whereas the VEET utilizes a wider beam and generates an 8 × 8 array of distance measurements that are averaged during post-processing[21,27]. The finer resolution of the VEET likely improves the ability to detect targets across a broader visual field and over a wider range of working distances[21]. Furthermore, differing sampling rates could contribute to the discrepancies observed. Although datasets were resampled to 1-second intervals for alignment, this post-hoc adjustment inherently assumes conditions remain static between recording intervals. However, because the temporal gap for viewing distance is relatively narrow (e.g., VEET's 2 s vs. Clouclip's 5 s), its overall impact on cumulative daily distance quantification is likely minimal.

Sensor orientation may also contribute to the observed differences. The Clouclip features a 10° downward tilt[27], whereas the VEET is designed with a 20° downward tilt[21]. For both devices, the downward orientation aids in capturing near work material that is viewed with the eyes in downgaze, which is important because viewing angle fluctuates dynamically across individuals and tasks and rarely aligns with the horizontal plane of the spectacles. Previous work in our lab demonstrated that viewing angle during near work often greater than 10°[31]. Consequently, the VEET geometry may better align with habitual visual behaviors during reading and handheld

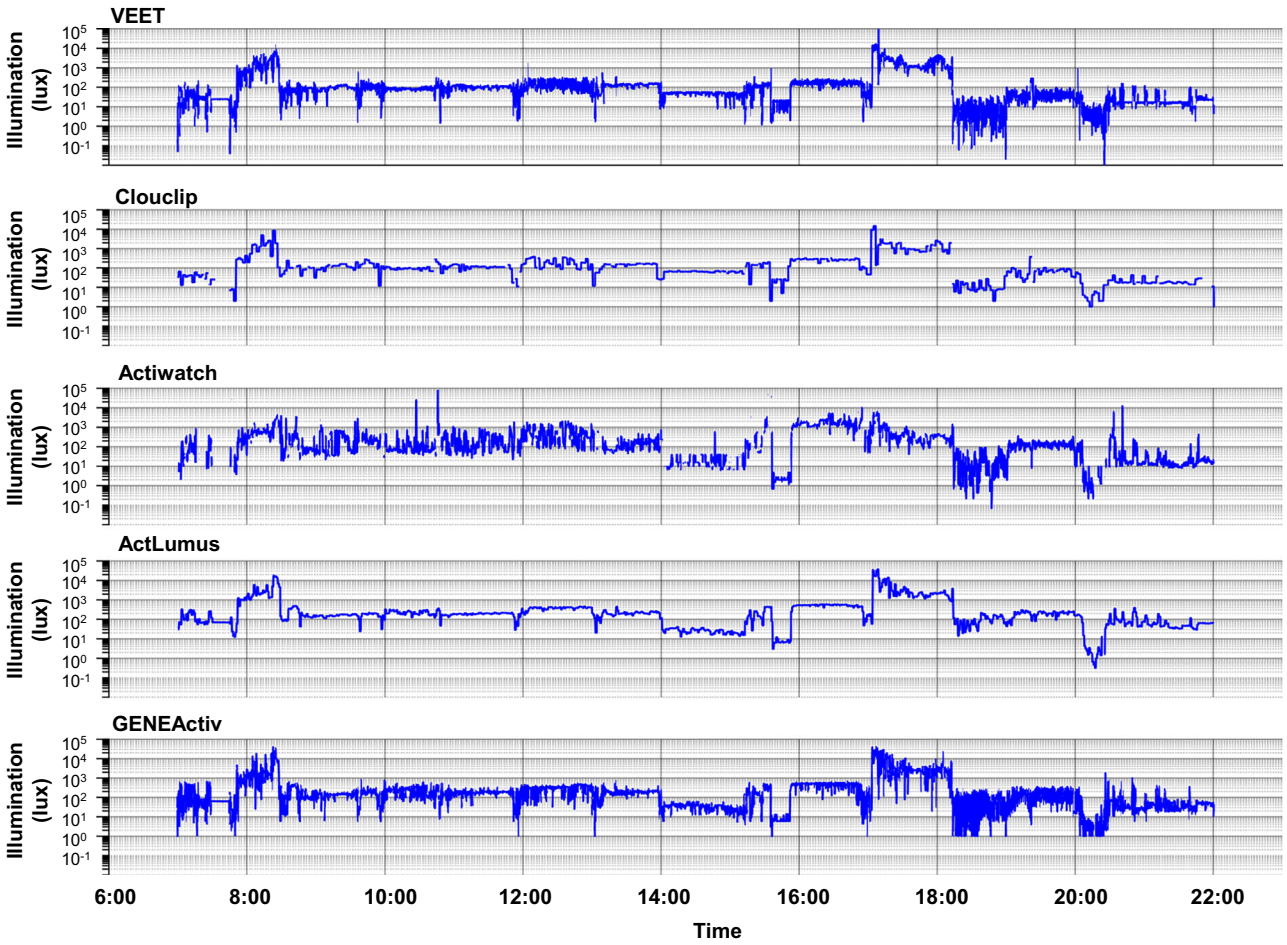


Fig. 4 | Representative ambient illumination from one participant recorded over the observation period across the five sensors. All datasets were resampled to a 1-second interval. The representative participant was selected randomly from the subset of participants who maintained 100% complete data capture and optimal signal quality (zero missing data, wear-time non-compliance, or sensor artifacts) across all wearable devices simultaneously.

device use. In contrast, the narrower field and shallower tilt of the Clouclip may increase the likelihood of missed detections during near viewing. The greater far viewing duration for the Clouclip is likely attributable to these missed detections, which generated error codes and were binned into the far viewing category during data processing.

Despite these differences in absolute viewing durations, the VEET and Clouclip demonstrated moderate-to-strong correlations across near, intermediate, and far viewing categories, indicating that both devices capture similar overall behavioural trends. Agreement was strongest for far viewing and somewhat lower for intermediate viewing, suggesting that transitions between working distances may be particularly sensitive to differences in sensor geometry and sampling characteristics. Collectively, these findings support the utility of spectacle-mounted ToF sensors for objective monitoring of near work, while also emphasizing that output from different devices should not be considered directly interchangeable without calibration or cross-validation.

For ambient illumination, all five sensors demonstrated strong agreement in the highest illumination category (≥ 1000 lux), corresponding to bright outdoor light exposure. This consistency suggests that all evaluated devices, whether spectacle-mounted or wrist-worn, are similarly capable of quantifying broad outdoor exposures, a clinically relevant metric in myopia research[32]. However, systematic differences emerged for indoor environments. The Actiwatch recorded substantially greater durations in scotopic illumination (< 3 lux) and proportionally less duration in photopic indoor illumination (30 to < 1000 lux) compared to the other four sensors. Inspection of the raw data demonstrated frequent zero-lux values from the Actiwatch during periods in which the other sensors measured low, but non-zero, illumination. These findings suggest differences in low-light sensitivity thresholds, cosine correction[33], or baseline calibration of the Actiwatch relative to the other devices. Importantly, such discrepancies would likely remain undetected if analyses were restricted to simple indoor-versus-outdoor classifications, as is common in the literature[21,27,34]. These four intensity bins were selected for their physiological relevance to myopia research. This granular analysis of absolute lux levels made the differences in low-light handling between devices evident. Furthermore, the disparity in sampling rates across the five sensors (1–120 s) could contribute to some extent to the observed variations. Because ambient illumination generally remains stable within a given environment, with rapid shifts typically occurring only during transitions, such as moving from indoors to outdoors, devices with lower sampling frequencies (e.g., 120 s) can still accurately characterize the majority of cumulative daily light exposure. While these missed transient shifts accumulate over the day, they likely account for only a modest portion of the total differences. Because these observations are derived from real-world scenarios, further confirmation of hardware-specific discrepancies within controlled, laboratory-based settings is warranted.

A secondary aim of this study was to compare objective measurements with self-reported activity logs. Although activity logs reduce some of the recall bias associated with retrospective questionnaires[35–37], significant discrepancies were demonstrated between subjective and objective measures. Here, activity logs underestimated near-viewing duration and overestimated intermediate-viewing duration relative to the wearable sensors,

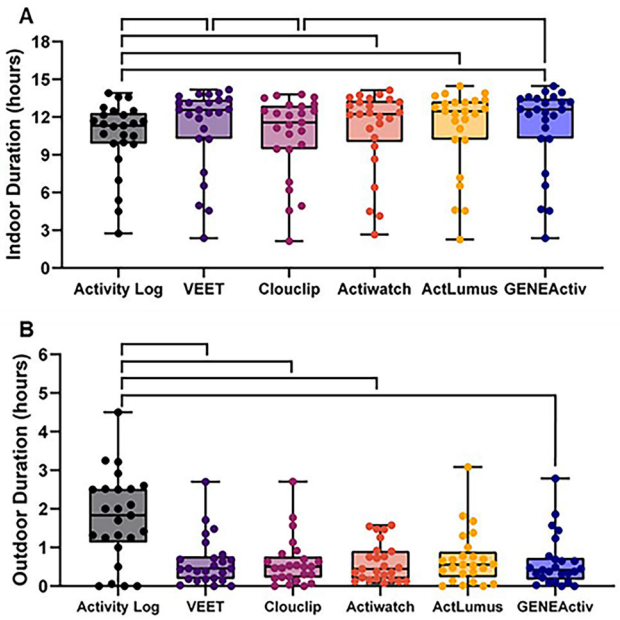


Fig. 5 | Duration of (A) **indoor (< 1000 lux) and** (B) **outdoor (≥ 1000 lux) illumination for the activity log and five sensors ($N = 25$).** Middle horizontal lines inside the boxes represent the median, and the upper and lower boundaries of the boxes represent the interquartile range. Whiskers represent the minimum and maximum values (range), with all individual data points superimposed as filled circles. Brackets indicate a statistically significant difference ($p < 0.05$) between pairs.

suggesting that assigning a single viewing-distance category to an activity does not adequately capture the dynamic nature of real-world visual behavior. Even within a single task, viewing distance may fluctuate continuously over time, which cannot be captured through subjective measures.

Differences were also observed for ambient illumination. Activities categorized subjectively as "outdoors" often produced illumination levels below the 1000 lux threshold, required for classification as outdoor exposure, by the sensors. This threshold has been adopted in myopia research to distinguish dimmer indoor conditions from brighter outdoor conditions. This discrepancy may reflect situations such as driving, sitting under covered structures, or remaining near windows, where participants perceived themselves to be in a particular location that might not have corresponded to the absolute threshold of 1000 lux. These findings further demonstrate the limitations of subjective classification approaches for quantifying environmental exposures relevant to myopia research. Nevertheless, activity logs remain valuable when utilized in conjunction with objective data, providing essential context and yielding a more comprehensive understanding of visual behavior. This subjective + objective hybrid approach might represent an effective strategy for future myopia research[20].

Strengths of this study include the simultaneous comparison of five wearable sensors under real-world conditions and the integration of both objective and subjective behavioral measures. This study provides a direct comparison of a broad range of currently available devices within the same cohort. The study also benefited from excellent participant compliance despite the burden of simultaneously wearing multiple sensors and maintaining detailed activity logs.

Some limitations should be taken into consideration. Despite careful configuration before dispensing the devices, some data loss occurred due to hardware malfunctions and user errors, such as incomplete charging of the spectacle-mounted devices, reducing the final sample size. Additionally, data collection was limited to one season (summer) and involved primarily young adults in a university setting. Environmental exposures and behavioral patterns may differ in children and across seasons. Furthermore, the Actiwatch's limited sensitivity in low-light environments (< 3 lux) should be carefully considered, as this has clear implications for studies focusing on nocturnal behaviors or dim light exposure. Additionally, because both the VEET and Clouclip lack direct pupillary gaze tracking, they provide an estimate rather than an exact measurement of viewing distance, as their fixed downward tilt cannot fully account for dynamic shifts in visual fixation[24]. Finally, a fundamental limitation of evaluating these sensors under real-world, habitual conditions is the absence of a gold-standard reference measurement. Although this study evaluated device agreement under habitual conditions, controlled laboratory validation against a true gold standard across a wider range of viewing distances and illumination conditions would further clarify the sources of disagreement observed here.

In conclusion, wearable sensors provide valuable objective measurements of visual environment and behavior, but differences exist across currently available devices. The VEET and Clouclip demonstrated differing viewing-distance profiles, likely related to sensor geometry and detection strategy, whereas all five sensors showed strong agreement for bright outdoor light exposure. Subjective activity logs provided useful contextual information but were significantly different than objectively measured viewing distance and ambient illumination. These findings highlight the importance of careful device selection and standardized analytical approaches in future work.

Methods

Participants and protocol

Healthy adults aged 18 years and older were recruited between May 2025 and October 2025 via institutional emails, printed flyers posted across university campuses, and word-of-mouth. All data collection took place in Houston, Texas. Written informed consent was obtained from all participants following a comprehensive explanation of the study procedures and objectives. Exclusion criteria were best-corrected visual acuity worse than 20/25 or any ocular or systemic medical condition causing an inability or unwillingness to wear the sensors. The protocol was approved by the Institutional Review Board at the University of Houston, and all procedures adhered to the tenets of the Declaration of Helsinki.

An initial screening visit was conducted to determine baseline eligibility, during which visual acuity and refractive error were measured. Objective noncycloplegic refraction was obtained using a WAM-5500 (Grand Seiko, Japan). Five measurements were collected for each eye as participants viewed a distance target through the open window of the instrument, then subsequently averaged. Spherical equivalent refraction (SER) was calculated and utilized to categorize participants as either non-myopic (SER > −0.50 D) or myopic (SER ≤ −0.50 D).

Following screening, eligible participants were instructed to wear five sensors simultaneously, comprising two spectacle-mounted and three wrist-worn sensors (described below), for one full day during waking hours, for a minimum of 14 h. Spectacle-mounted sensors were positioned on plano lens spectacle frames. Myopic participants were encouraged to wear contact lenses for the experimental day. If preferred, the frames with the sensors were fitted with prescription lenses for myopic participants.

Participants were permitted to select any day of the week for data collection. Concurrently, participants were instructed to maintain a detailed activity log to document their daily activities throughout the data collection period. Specifically, participants recorded the start and end times of the sensor wear period, as well as distinct tasks with brief descriptions of the activities performed (e.g., reading, driving, or household chores).

Sensors

Five wearable sensors were evaluated, two spectacle-mounted devices, the Visual Environment Evaluation Tool (VEET, Meta LLC, United States) and the Clouclip (Glasson Technology Co., Ltd, China), and three wrist-worn devices, the Actiwatch (Philips Respironics, United States), ActLumus (Condor Instruments, Brazil), and GeneActiv (ActivInsights, Canada) (Figs. 7, 8). The specifications for each device are shown in Table 5. Prior to dispensing the devices, each sensor was configured using its respective proprietary software.

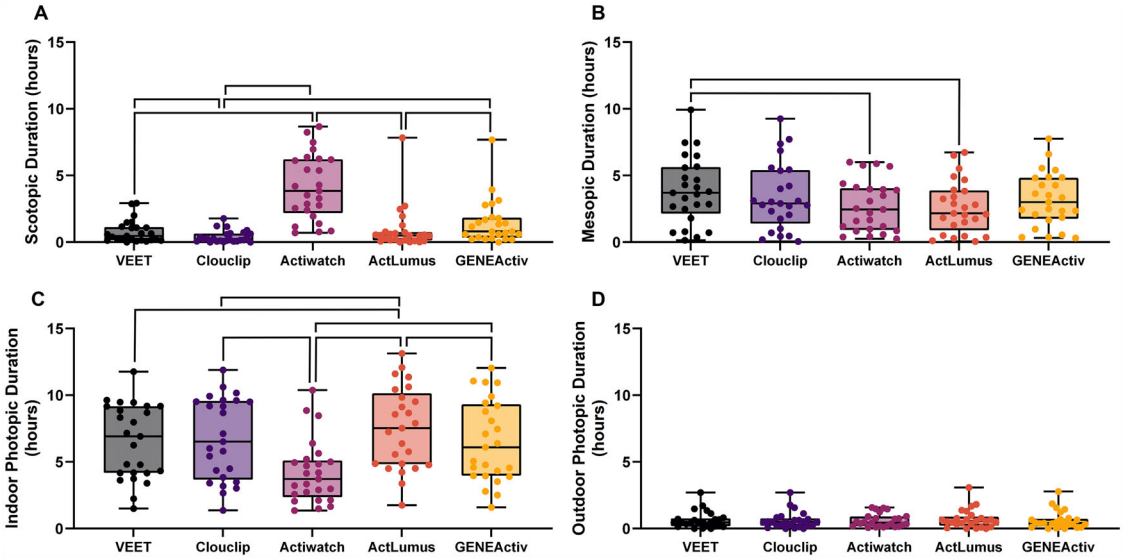


Fig. 6 | Ambient light exposure durations across five sensors ($N = 25$), categorized into four illuminance bins, (A) **scotopic (< 3 lux),** (B) **mesopic (3 to < 30 lux),** (C) **indoor photopic (30 to < 1000 lux), and** (D) **outdoor photopic (≥ 1000 lux).** Middle horizontal lines inside the boxes represent the median, and the upper and lower boundaries of the boxes represent the interquartile range. Whiskers represent the minimum and maximum values (range), with all individual data points superimposed as filled circles. Brackets indicate a statistically significant difference ($p < 0.05$) between pairs.



Fig. 7 | The five wearable sensors utilized in this study. A Visual Environment Evaluation Tool (VEET), (**B**) Clouclip, (**C**) Actiwatch, (**D**) ActLumus, and (**E**) GENEActiv. The Clouclip was mounted in alignment with the VEET on the right temple arm of the spectacle frame for data collection.

The VEET consists of integrated sensors on both the right and left temples, each with identical components, including an infrared ToF rangefinder, a light sensor, a spectrometer, and a gyroscope. The rangefinders have a reported range of 5-400 cm. The right ToF rangefinder was designed to capture near viewing and is angled 20° downward and 4° nasal, while the left ToF rangefinder is directed straight ahead (0°). For this study, only the right temple data were considered, given its optimal tilt for capturing near-viewing behaviors. The light sensor uses three channels, ultraviolet, photopic, and infrared, and estimates ambient illumination in lux, with a reported range of 0–128,000 lux. The VEET also includes nine spectral channels from the ultraviolet to infrared range; however, spectral data were not assessed in the current study. The VEET was initialized using the VEET Manager software, set to a 2-second sampling epoch for both distance and illumination measurements.

The Clouclip is a spectacle-mounted ToF rangefinder and light sensor. The infrared tracking beam is directed 10° downward and has measurement range 5–120 cm for distance and 1 to 65,336 lux for ambient illuminance. Distance is captured every 5 s, and ambient illuminance is captured every 2 min. The Clouclip also contains an angular acceleration sensor (X, Y, Z axes) to detect nonwear. If the angular acceleration sensor is unchanged for more than 40 s, the device goes to sleep for a minimum of two minutes and until movement is detected. The Clouclip was aligned and mounted alongside the VEET on the right temple of the spectacle frame. Participants wore the VEET–Clouclip–paired spectacles fitted with either plano or prescription lenses, as appropriate.

The Actiwatch is a wrist-worn device that records ambient illuminance and activity at 32 Hz. Ambient illuminance measurement range is 0.1–35,000 lux for broadband light. Activity is recorded via an in-built accelerometer and is expressed in counts per minute. The Actiwatch was configured via Actiware software with a 15-second sampling epoch.

The ActLumus is a wrist-worn device that captures ambient illuminance, with a range of 1–100,000 lux, as well as spectral composition (10-channels), activity, and temperature. Apart from recording ambient illumination, the device is also used to monitor body movements and sleep quality. The ActLumus was configured via ActStudio with a 60-second epoch.

The GENEActiv is primarily worn on wrist but can also be worn on other body parts. The GENEActiv captures ambient illuminance, with a range of 0 to 20,000 lux, as well as activity and temperature. Similar to the ActLumus, the GENEActiv is designed to also monitor activity and sleep quality. The GENEActiv was configured via GENEActiv software with a 1-second epoch.

Following data collection, the sensors were retrieved, and the data were extracted. Data from the Actiwatch, ActLumus, GeneActiv, and VEET were extracted locally using their corresponding software. Clouclip data, which are synchronized via a mobile application and automatically uploaded to a cloud platform, were exported as comma-separated values (CSV) files using secure login credentials.

Analysis

Data processing, statistical analyses, and visualizations were performed using MATLAB R2025b (MathWorks, Natick, MA). Following data extraction, device-specific pre-processing was conducted. For the VEET, the raw ToF distance output consists of 64 distinct spatial zones, each accompanied by a signal-to-noise confidence value. Distance values with low

confidence (< 10%) or reported as zeros were flagged invalid and converted to not-a-number (NaN), as previously described in ref. 21. Valid data across the 64 zones were then spatially averaged to yield a single distance value. This specific filtering step was not applicable to the other devices because they output a single value per timestamp.

The VEET ToF and gyroscope data were used to determine a standardized wear time across all sensors. The gyroscope provides an inertial measurement unit (IMU), a measure of activity, which established whether the spectacles were being worn. The VEET data were initially trimmed to exclude leading and trailing inactive periods, characterized by invariant ToF and IMU data values prior to and following the active wear period. Any period of non-wear during the wear period was detected using a dual-sensor (ToF and IMU) approach. This logic distinguishes true device non-wear from stationary viewing behaviors (e.g., remaining still while watching television). The final analytical dataset was determined by computing the valid wear time, which was then applied to the datasets from the other four devices. This process ensured that data corresponding to non-wear periods were universally removed, generating a clean, synchronized dataset representing true valid wear time across all devices.

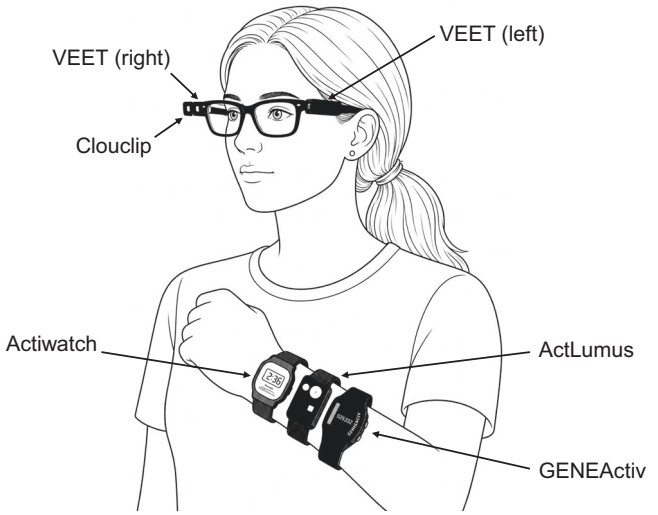


Fig. 8 | Positioning of the sensors. The VEET is integrated into the right and left temples, the Clouclip was attached to the right temple, and the Actiwatch, ActLumus, and GENEActiv were worn on the left wrist. Figure rendered with the assistance of ChatGPT (OpenAI).

Given the heterogeneity in sampling intervals across the five devices, the refined valid datasets were temporally aligned prior to statistical comparison. For each participant, all device datasets were aligned to a common start and end time and resampled to a uniform 1-second interval using custom MATLAB (MathWorks, Natick, MA) programs. Value propagation between consecutive sensor readings was implemented using a stepwise forward-repeat algorithm (Last Observation Carried Forward [LOCF]), with a maximum carry-forward threshold of 60 seconds (for distance) and 120 s (for illumination); gaps exceeding these thresholds and identified non-wear periods were treated as missing data (NaN) and excluded prior to duration calculations. Conversely, pre-existing sensor NaN values inherent to device logging (e.g., Clouclip readings beyond its 100 cm detection range) were preserved and retained, as they represented valid distance-viewing events rather than data loss.

Data were stratified by the environmental variables of interest, distance and illumination. For distance measures, ToF distance data from the VEET and Clouclip were categorized into three ranges: near (10 to < 60 cm), intermediate (60 to < 100 cm), and far (≥ 100 cm). The cumulative daily duration (in hours) spent in each category was calculated, and the predictive agreement between the two devices was assessed using linear regression and correlation coefficients.

Table 5 | Technical specifications of the five sensors utilized in this study.

	VEET	Clouclip	Actiwatch	ActLumus	GENEActiv
Model/version	v2.0	M2	Spectrum Plus	AL0101	v1.2
Dimensions (L x W x H)	15 × 1.5 × 2.0 cm	4.5 × 1.3 × 0.8 cm	4.8 × 3.7 × 1.5 cm	3.4 × 2.7 × 1.3 cm	4.3 × 4.0 × 1.3 cm
Weight	60 g (with frame and plano lenses)	5 g (without glasses)	31 g (with wristband)	31 g (with wristband)	26 g (with wristband)
Location	Spectacle-integrated, both right and left temples	Spectacle-mounted, right temple	Wrist	Wrist/neck/clipped to clothing	Wrist/clipped to clothing
Battery life*	1–6 days	1 day	60 days	30 days	60 days
Sensors	Time of flight (5–400 cm), accelerometer, ambient light (0–128,000 lux), temperature, spectral channels (11 channels from UV to infrared)	Time of flight (5–120 cm), ambient light (1–65,000 lux)	Accelerometer, ambient light (1–35,000 lux), spectral channels (red, green, blue)	Accelerometer, ambient light (1–100,000 lux), temperature, spectral channels (10 channels from UV to infrared)	Accelerometer, ambient light (0–20,000 lux), temperature
Parameters measured	distance and illumination	distance and illumination	illumination	illumination	illumination
Communication	USB	Bluetooth	miniUSB	USB	USB
Sampling rate	Up to 0.5 Hz	0.2 Hz for distance 0.0083 Hz for illumination	32 Hz	0.017 Hz	10–100 Hz
Epoch length	2 s for both distance and illumination§	5 s for distance 2 mins for illumination	15 s for illumination§	60 s for illumination§	1 s for illumination§
Memory	29 GB	can store data for up to 8 days	32 MB	8 MB	1 GB

*Battery life varies with device configuration.
§Specific epoch length configured for this study.

For ambient illumination, data from each sensor were binned into four intensity categories: scotopic (< 3 lux), mesopic (3 to < 30 lux), indoor photopic (30 to < 1000 lux), and outdoor photopic (≥ 1000 lux). These four intensity bins were selected for their physiological relevance to myopia research, as they effectively categorize visually distinct environmental lighting conditions. The total daily duration (in hours) recorded within each category was computed for each device for comparison across devices.

The self-reported activities from the log were systematically categorized into distance (near, intermediate, far, or a combination) and environmental (indoors or outdoors) domains. Following this categorization, the total duration for each category was computed to facilitate direct comparisons with the objective sensor data.

Statistical analysis

Data normality was evaluated using the Kolmogorov-Smirnov test. Due to non-normal distributions of key continuous variables (wear time, viewing distance, and light duration), data are presented as median (interquartile range, [IQR: Q1, Q3]). Differences in wear time were evaluated using the non-parametric Wilcoxon signed-rank test. The non-parametric Friedman test was utilized to compare viewing distance and ambient illumination durations across the wearable sensors and subjective activity logs. Effect sizes for omnibus Friedman tests were calculated using Kendall's coefficient of concordance (W), interpreted as small (0.1), moderate (0.3), and large (≥ 0.5). Significant main effects were followed by post-hoc pairwise comparisons using the Dunn-Bonferroni test. To avoid inflation of Type II error, this multiplicity correction was applied strictly within each individual Friedman analysis adjusting only for the number of pairwise comparisons within a specific environmental category rather than across the broader family of all study outcomes. In addition, agreement between devices was evaluated using linear regression and Bland–Altman analysis, reporting mean bias and 95% limits of agreement. Statistical significance was defined as $p < 0.05$.

Although this study was an exploratory comparison of wearable sensors, a sample size of 25–28 participants provided adequate precision and power for this within-subject design. Specifically, $N = 25$ allows estimation of the mean bias between devices to within ± 0.41 standard deviations with 95% confidence. It also provides 80% statistical power ($\alpha = 0.05$) to detect moderate-to-large within-subject effects (Cohen's d ≥ 0.56), which was sufficient given the consistently large effect sizes observed across our comparisons.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ongoing and planned analyses but are available from the corresponding author on reasonable request.

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Author contributions

R.P.S.: conceptualization; data curation, formal analysis, investigation, methodology, software, validation, visualization, writing – original draft, writing – review & editing. N.Y.: data curation, methodology, validation, visualization, writing – review & editing. P.N.: data curation, methodology, validation, visualization, writing – review & editing. L.A.O.: conceptualization; data curation, investigation, methodology, supervision, validation, visualization, writing – original draft, writing – review & editing.

Competing interests

LAO: Meta, LLC (research support), Zeiss (consulting), US Patent (US 11375890 B2).

Additional information

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